



CS2009

Spring 2024

Assignment# 04

Design and Analysis of Algorithm

SUBMISSION GUIDELINES:

1. The assignment must be submitted within the due date.
2. The assignment must be on paper and in handwritten form.
3. **Plagiarism is not allowed. In the case of a pledge, you will score zero marks.**
4. Do not submit your assignment after the deadline.
5. Do not copy from any other source.
6. It is an individual submission.

Question no: 1

- Consider a B-tree of order 5. Perform a dry run of inserting the following Alphabets in sequence: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z. Draw the B-tree after each insertion.
- Using the final B-tree, perform a dry run of deleting the following Alphabets in sequence: L, F, P and V. Draw the B-tree after each deletion.

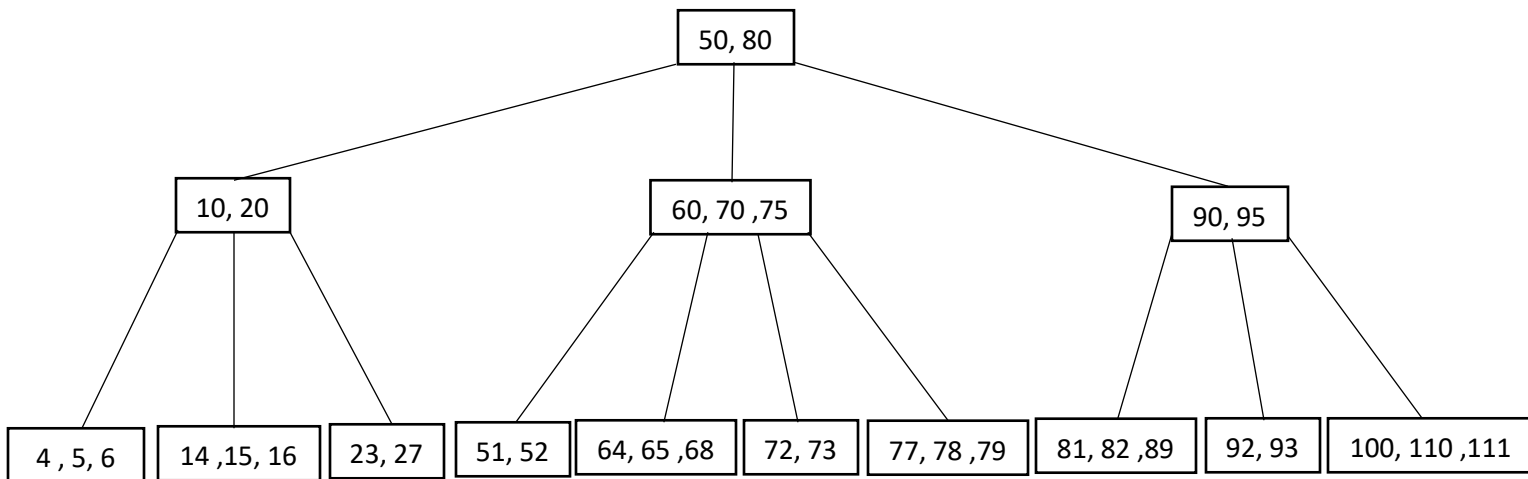
Question no: 2

- Consider a B-tree of order 5. Perform a dry run of inserting the following Numbers in sequence: 3, 7, 9, 23, 45, 1, 5, 14, 25, 24, 13, 11, 8, 19, 4, 31, 35, 56, 2, 6, 12. Draw the B-tree after each insertion.
- Using the final B-tree, perform a dry run of deleting the following Numbers in sequence: 4, 5, 7 and 14. Draw the B-tree after each deletion.

Question no: 3

- Consider a B-tree of order 3. Perform a dry run of inserting the following Numbers in sequence: 3, 7, 9, 23, 45, 1, 5, 14, 25, 24, 13, 11, 8, 19, 4, 31, 35, 56. Draw the B-tree after each insertion.
- Using the final B-tree, perform a dry run of deleting the following Numbers in sequence: 1, 13, 23, 4 and 19. Draw the B-tree after each deletion.

Question no: 4



Using the final B-tree of order-5, perform a dry run of deleting the following Numbers in sequence: 64, 23, 72, 65, 20, 70, 95, 77, 80, 100, 6, 27, 60, 16 and 50. Draw the B-tree after each deletion.

Question no: 5

- Consider a B+ tree of order 5. Perform a dry run of inserting the following Alphabets in sequence: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z. Draw the B+ tree after each insertion.
- Using the final B+ tree, perform a dry run of deleting the following Alphabets in sequence: L, F, P and V. Draw the B+ tree after each deletion.

Question no: 6

- Consider a B+ tree of order 5. Perform a dry run of inserting the following Numbers in sequence: 3, 7, 9, 23, 45, 1, 5, 14, 25, 24, 13, 11, 8, 19, 4, 31, 35, 56, 2, 6, 12. Draw the B+ tree after each insertion.
- Using the final B+ tree, perform a dry run of deleting the following Numbers in sequence: 4, 5, 7 and 14. Draw the B+ tree after each deletion.

Question no: 7

- Consider a B+ tree of order 3. Perform a dry run of inserting the following Numbers in sequence: 3, 7, 9, 23, 45, 1, 5, 14, 25, 24, 13, 11, 8, 19, 4, 31, 35, 56. Draw the B+ tree after each insertion.
- Using the final B+ tree, perform a dry run of deleting the following Numbers in sequence: 1, 13, 23, 4 and 19. Draw the B+ tree after each deletion.

Question no: 8

You are a hiker preparing for a trip into the wilderness, and you have a knapsack with a maximum weight capacity of 10 kilograms. You are presented with the following items:

- Item 1: Weight = 4 kg, Value = \$8
- Item 2: Weight = 3 kg, Value = \$6
- Item 3: Weight = 6 kg, Value = \$12
- Item 4: Weight = 2 kg, Value = \$5
- Item 5: Weight = 5 kg, Value = \$10

Using dynamic programming, determine the maximum total value you can achieve by selecting a subset of the given items to pack into your knapsack, considering the weight constraint of 10 kilograms. Show your step-by-step decision-making process, including the construction of the dynamic programming table, and indicate which items you choose to include in your knapsack to maximize the total value.

Highlight each cell where the memorization perform and also identify picked items for getting maximum profit in knapsack using backtracking.

Question no: 9

Explain the concept of fractional knapsack and solve the above example of knapsack using a fractional knapsack.